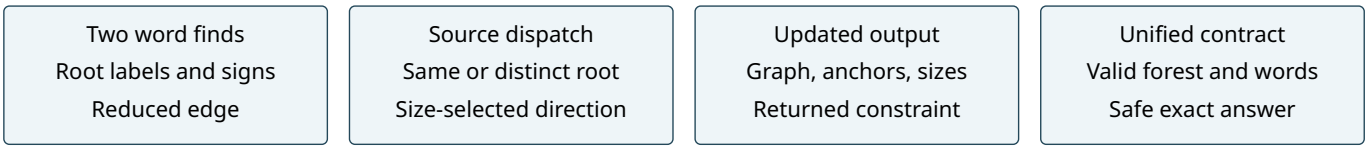
Unified signed edge dispatcher after two word finds

Producer T-1346; archive T-1347. The model follows the same-root and distinct-root branches of SignedGraph::edge at 4cfb6c539c. Its output includes graph/anchor state, the size vector and the returned constraint. Distinct roots use the native less-than-or-equal size comparison; equal sizes therefore use the forward orientation.



Result. The unified after_finds_contract composes the published same-root, union-state, size-selection and returned-constraint lemmas. Under their explicit input assumptions, both dispatch branches preserve canonical potentials and anchors, the signed graph/anchor invariant, an existential forest-rank certificate and a safe return. Seven named declarations compile, including canonical and exact target-word equality consequences and exclusion of the inconsistent-result branch.

Sizes. Same-root dispatch leaves the size vector unchanged. Distinct-root dispatch updates the selected parent by adding the selected child size. The exact integer sum theorem assumes disjoint finite component sets, equality of their cardinalities to the initial root sizes, and n < USize.size. It proves the selected parent contains the sum under that bound. These cardinality premises are explicit; this theorem does not silently identify arbitrary stored sizes with actual graph components.

Checks. Seven controls cover a redundant cycle, an inconsistent cycle, a cycle returning six modulo seven, positive and negative new anchors, tied-size forward parenting and reversed parenting. The two size-orientation controls state a 64-bit platform premise. Fourteen separate named axiom readbacks contain only propext, Classical.choice and Quot.sound. Three malformed inventories are rejected. Lean rejects a false claim that the inconsistent same-root example returns no constraint. The first main-proof attempt failed before explicit unfolding of the same-root dispatch and is retained.

Assumptions. Prime UInt64 modulus, 2 < r <= 2^63, canonical input coefficients and anchors, signs +1 or -1, valid input graph/anchor equations, a forest certificate, the two stated word-find runs and a true incoming signed edge. Exact word equality also requires a canonical target word. The construction composes mathematical models; it does not derive the incoming relation from curve geometry or provide a Rust compiler refinement.

Evidence boundary. All module checks use one thread, 4096 MiB and a 120-second bound. Source and imported dependencies are hash-bound before and after checks; compiled dependencies are reused, with no independent binary/kernel recheck. Pre/post hashes cannot exclude transient concurrent edits. Independent source-fidelity review remains pending. The dispatcher carries the size update, but full component-owner/cardinality preservation across all branches remains a separate composition obligation. Native counters, allocation and partial-failure behavior are not modeled here.

Next construction. Add source counter increments and their overflow bounds, then connect the complete dispatcher to the component-owner/root-size invariant and native storage representation. No scientific execution, cost measurement, cohort change or budget renewal occurred. The full 1,260-target obligations and independent-review requirements remain unchanged.